IERG4320/IEMS5726 — Chapter 2: Basic Statistical Test — Cheat Sheet

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Key Terminology (基础概念)

- **Population** (总体): all data you want to study
  比如"全校所有学生"
- **Sample** (样本): a portion you actually collect
  比如"随机抽了100个学生"
- **Variable** (变量): anything you can measure
  比如身高、成绩、年龄
- **Descriptive stats** (描述统计): describe what data looks like
  比如"平均分是75"
- **Inferential stats** (推断统计): use sample to predict population
  用100人的数据推测全校情况

Central Tendency (集中趋势)

- **Mean** (均值/平均数): $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
  所有数加起来除以个数
- **Median** (中位数): middle value when sorted
  数据排好序后最中间那个
- **Mode** (众数): most frequent value
  出现次数最多的那个值

Spread (离散程度)

- **Range** (范围): $\max - \min$
- **Variance** (方差): $\sigma^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$
  每个值离均值有多远，取平方再平均
- **Std Dev** (标准差): $\sigma = \sqrt{\sigma^2}$
  方差开根号，更直观看数据有多分散

Hypothesis Testing (假设检验)

核心问题：观察到的结果到底是真的有意义，还是碰巧?

> **Goal** (目的): Rule out "luck" as the explanation.
> 排除"纯属巧合"这种可能性。

6 Steps (六步走)

1. **State Hypotheses** (提出假设)
   - H_0 (零假设): no effect (没效果)
   - H_a (备择假设): effect exists (有效果)
2. **Choose** α (选显著性水平): 0.05, 0.01, or 0.1
   $\alpha = 0.05$意思是：巧合概率<5%时，我就认为不是巧合
3. **Pick a test** (选检验方法): Z, t, χ^2, etc.
4. **Calculate test statistic** (算统计量)
5. **Find critical region** (确定拒绝域)
6. **Decision** (做决定): reject H_0 or not

Critical Regions (拒绝域)

One-tailed (单尾，只看一个方向):

- Left: reject H_0 if $z < -z_\alpha$
- Right: reject H_0 if $z > z_\alpha$

Two-tailed (双尾，两边都看): reject if $|z| > z_{\alpha/2}$
比如检验硬币公不公平，正面太多或太少都算有问题

p-value (p值)

> p-value = probability of seeing this result (or more extreme) when H_0 is true.
> p值 = 假如真的没效果，出现当前结果的概率有多大
>
> **Rule**: $p < \alpha \Rightarrow$ reject H_0 (拒绝零假设 = 有显著效果)
> p值越小，越能说明"不是巧合"

Example (例子): Test if coin is fair ($H_0 : p = 0.5$), flip 100 times, get 60 heads:
$p = 2\,P(X \geq 60) = 0.0574 > 0.05$
$\Rightarrow$ Cannot reject H_0 (不能说硬币不公平)

Fisher's Exact Test (Fisher精确检验)

适用场景：2×2表格、小样本、变量只有两类（是/否）

- For **2×2 contingency tables** (列联表)
- Tests if two binary variables are independent (两个变量是否有关)
- Computes **exact** probability (精确计算，不是近似)

Contingency Table (列联表)

	Group I	Group II	Total
Category A	a	b	$a+b$
Category B	c	d	$c+d$
Total	$a+c$	$b+d$	n

Example (例子：骰子公不公平？)

Rolling 2 dice	Exper.	Theory
Sum ≥ 10	15	6
Sum < 10	57	30

```python
import scipy.stats as stats
oddsratio, pvalue = stats.fisher_exact(
    [[15, 57], [6, 30]], "greater")
print(oddsratio, " ", pvalue)
# 1.3158  0.4051
```

$p = 0.405 \gg 0.05 \Rightarrow$ Cannot reject H_0
p值远大于0.05，没有证据说骰子不公平

Pros & Cons (优缺点)

优点：小样本精确；缺点：只能处理2×2表格，大数据慢

Chi-squared Test (卡方检验, χ^2)

适用场景：多类别频率比较，看实际频率和预期频率是否有显著差异

Test Statistic (公式)

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- O_i: observed count (实际观察到的次数)
- $E_i = Np_i$: expected count (理论上期望的次数)
- $df = k - 1$ (自由度 = 类别数 − 1)

核心思想：χ^2越大，说明实际和预期差得越多

Chi-square Distribution (卡方分布)

If $X_1, \ldots, X_k \sim N(0,1)$ independent, then:

$$Y = X_1^2 + X_2^2 + \cdots + X_k^2 \sim \chi^2(k)$$

$E[Y] = k, \quad \mathrm{Var}(Y) = 2k$

Example (例子：骰子有没有偏？)

骰子掷60次，各面出现：5, 8, 9, 8, 10, 20次。公平骰子每面应该10次。

i	1	2	3	4	5	6	Σ
O_i	5	8	9	8	10	20	60
E_i	10	10	10	10	10	10	60
$\frac{(O-E)^2}{E}$	2.5	0.4	0.1	0.4	0	10	**13.4**

注意：6出现了20次，远超预期，是主要偏差来源

```python
from scipy.stats import chisquare
chi2, pvalue = chisquare([5,8,9,8,10,20])
print(chi2, " ", pvalue)
# 13.4  0.01991
```

Decision ($df = 5$, $\chi^2 = 13.4$):

- 95%: $13.4 > 11.07 \Rightarrow$ **Reject** H_0 (骰子不公平!)
- 99%: $13.4 < 15.09 \Rightarrow$ Cannot reject (没法下结论)

同样的数据，标准不同结论就不同，这就是α的意义

Wilcoxon Rank-Sum Test (秩和检验)

(Mann–Whitney U Test)

适用场景：两组独立数据的比较，数据不需要正态分布（t检验的替代品）

Key Idea (核心思想)

- H_0: $P(X > Y) = P(Y > X)$ (两组没差别)
- Both groups must be independent (两组互相独立)

如果两组没差别，混合排名后两组的排名应该差不多

Steps (步骤)

1. Mix both groups, rank ascending (混一起从小到大排名)
   相同值取排名的平均
2. T = sum of ranks of one group (一组的排名加起来)
3. $U = T - \frac{n(n+1)}{2}$ (算U统计量)
4. $\mu_U = \frac{n_1 n_2}{2}$, $\sigma_U = \sqrt{\frac{n_1 n_2 (n_1+n_2+1)}{12}}$
5. $z = \frac{U-\mu_U}{\sigma_U} \to$ get p-value

Example

X (15 values), Y (14 values):
$U_X = 152$, $\mu_U = 105$, $\sigma_U = 56.13$
$z = 0.837$, $p = 0.040$

在99%水平下不能拒绝H_0，没法说两组有差别

```python
from scipy.stats import ranksums, mannwhitneyu
import numpy as np
X = np.array([3.99, 3.79, 3.60, 3.73, 3.21,
    3.60, 4.08, 3.61, 3.83, 3.31,
    4.13, 3.26, 3.54, 3.51, 2.71])
Y = np.array([3.18, 2.84, 2.90, 3.27, 3.85,
    3.52, 3.23, 2.76, 3.60, 3.75,
    3.59, 3.63, 2.38, 2.34])

# Method 1: ranksums
U, pvalue = ranksums(X, Y)
print(U, pvalue)  # 2.05  0.0402

# Method 2: mannwhitneyu
stat, pvalue = mannwhitneyu(X, Y)
print(stat, pvalue)  # 152.0  0.0423
```

两个函数结果略不同但结论一致

Wilcoxon Signed-Rank Test (符号秩检验)

适用场景：配对数据（同一组人的前后对比），数据不需要正态分布（配对t检验的替代品）

Key Idea (核心思想)

- H_0: paired samples come from same distribution
  两组配对数据来自相同分布（处理没效果）
- Data must be paired (数据必须配对，比如同一个人的前/后)

如果处理没效果，正的差值和负的差值应该差不多抵消

Steps (步骤)

1. $d_i = x_i - y_i$ (算每对的差值)
2. Drop pairs where $d_i = 0$ (差为0的扔掉)
3. Rank by $|d_i|$ ascending (按差值绝对值排名)
4. $W = \sum$ signed ranks (加上正负号的排名求和)
5. $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}}$
6. $t = W/\sigma_W$, use t-dist ($df = n-1$)

Example

X	Y	$X-Y$	Rank
6.1	5.2	0.9	3.5
7.0	7.9	−0.9	3.5
8.2	3.9	4.3	10
7.6	4.7	2.9	7
6.5	5.3	1.2	5
8.4	5.4	3.0	8
6.9	4.2	2.7	6
6.7	6.1	0.6	2
7.4	3.8	3.6	9
5.8	6.3	−0.5	1

$W = 46$, $\sigma_W = 19.62$, $df = 9$, $p = 0.0438$

在99%水平下不能拒绝H_0

```python
from scipy.stats import wilcoxon
import numpy as np
X = np.array([6.1, 7.0, 8.2, 7.6, 6.5,
              8.4, 6.9, 6.7, 7.4, 5.8])
Y = np.array([5.2, 7.9, 3.9, 4.7, 5.3,
              5.4, 4.2, 6.1, 3.8, 6.3])
T, pvalue = wilcoxon(X, Y)
print(T, pvalue)  # 5.0  0.01953
```

注意：Python用T统计量（正/负秩和的较小值），和手算W不同，但结论一致

Quick Reference (速查表：用哪个检验？)

Test	When to Use (什么时候用)
Fisher's Exact	2×2 table, small n, binary (小样本，是/否变量)
χ^2 Test	Multi-category frequency (多类别频率比较)
Rank-Sum (U)	2 independent groups, non-normal (两组独立数据)
Signed-Rank	Paired data, non-normal (配对数据)

Core logic (核心套路):
Compute statistic $\to$ get p-value $\to$ if $p < \alpha$, reject H_0
算统计量 $\to$ 得到p值 $\to$ p值$< \alpha$就拒绝零假设 $\to$ 结果不是巧合!

Other Tests (其他检验，了解即可)

- **t-test** (t检验): compare two groups (need normal data)
- **ANOVA** (方差分析): compare >2 groups
- **Correlation** (相关分析): are two variables linearly related?
- **Regression** (回归): predict Y from X